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Slidable post with airbag for wheelchair occupants

Abstract

A vehicle includes a vehicle floor and a post supported by the vehicle floor. The post is translatable relative to the vehicle floor. The vehicle includes a support member mounted to the post. The support member is rotatable relative to the post. The vehicle includes an airbag supported by the support member.

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Background/Summary

BACKGROUND

(1) Passenger vehicles are not typically designed to accommodate wheelchairs. Therefore, accommodations for wheelchairs in vehicles are typically installed with aftermarket modifications to a production vehicle. For example, a vehicle may be modified with a lift or the like to load a wheelchair onto a vehicle. Production vehicles typically do not have the ability to accommodate the wheelchair of an occupant in a manner allowing the occupant to sit in the wheelchair in the vehicle, or if they do, significant aftermarket modification is required.

(2) Additionally, vehicles are equipped with airbags which act as supplemental restraints for occupants during certain vehicle impacts. The airbags are located at various fixed positions in occupant compartments of vehicles. Vehicles may include airbags supported on a dash, side air curtains mounted to the roof rails, side airbags mounted to seats, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a vehicle having a post at a first position and an airbag at an uninflated position.

(2) FIG. 2 is a perspective view of the vehicle having the post at a second position and the airbag at the uninflated position.

(3) FIG. 3 is a perspective view of a portion of the post with a support member.

(4) FIG. 4 is a perspective view of a portion of a track supporting the post.

(5) FIG. 5 is a perspective view of a vehicle having the post at the first position and the airbag at an inflated position.

(6) FIG. 6 is a perspective view of the vehicle having the post at the second position and the airbag at the inflated position.

(7) FIG. 7 is a block diagram of components of the vehicle.

(8) FIG. 8 is a flowchart of a method for controlling the post and the airbag.

DETAILED DESCRIPTION

(9) <Insert Claim Summary after Inventor Review>

(10) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a vehicle **20** having a vehicle floor **22** and a post **24** supported by the vehicle floor **22** is shown. The post **24** is translatable relative the vehicle floor **22**. The vehicle **20** includes a support member **26** mounted to the post **24**. The support member **26** is rotatable relative to the post **24**. The vehicle **20** includes an airbag **28** supported by the support member **26**.

(11) Translation of the post **24** relative to the vehicle floor **22** enables the airbag **28** to be repositioned within the vehicle **20**. For example, the post **24** may be translated to a first position to control kinematics of a front-facing occupant of the vehicle **20** and may be translated to a second position to control kinematics of a rear-facing occupant of the vehicle **20**.

(12) With reference to FIGS. 1 and 2, the vehicle **20** may be any suitable type of ground vehicle, e.g., a passenger or commercial automobile such as a sedan, a coupe, a truck, a sport utility, a crossover, a van, a minivan, a taxi, a bus, etc.

(13) The vehicle **20** defines a vehicle-longitudinal axis **A1** extending between a front vehicle end (not numbered) and a rear vehicle end (not numbered) of the vehicle **20**. The vehicle **20** defines a cross-vehicle axis **A2** extending cross-vehicle from one side to the other side of the vehicle **20**. The vehicle **20** defines a vehicle-vertical axis **A3**. The vehicle-longitudinal axis **A1**, the cross-vehicle axis **A2**, and the vehicle-vertical axis **A3** are perpendicular relative to each other.

(14) The vehicle **20** may include a vehicle body **29**. The vehicle body **29** includes body panels (not all numbered) partially defining an exterior of the vehicle **20**. The body panels may present a class-A surface, e.g., a finished surface exposed to view by a customer and free of unaesthetic blemishes and defects.

(15) The vehicle body **29** defines an occupant compartment **30** to house occupants of the vehicle **20**. The occupant compartment **30** may extend across the vehicle **20**, i.e., from one side to the other side of the vehicle **20**. The occupant compartment **30** includes a front end and a rear end with the front end being in front of the rear end during forward movement of the vehicle **20**.

(16) The vehicle floor **22** defines the lower boundary of the occupant compartment **30**. The vehicle floor **22** may extend from the front end of the occupant compartment **30** to the rear end of the occupant compartment **30**. The vehicle floor **22** may include upholstery, for example, carpet, and may have a class-A surface facing the occupant compartment **30**, i.e., a surface specifically manufactured to have a high quality, finished, aesthetic appearance free of blemishes.

(17) The occupant compartment **30** defines a wheelchair seating area **32**. The wheelchair seating area **32** is the area occupied by a wheelchair **34** and occupant of the wheelchair **34** when properly seated in the vehicle **20**. The wheelchair **34** in the wheelchair seating area **32** may face the front end of the occupant compartment **30**, as shown in FIG. 1, or may face the rear end of the occupant compartment **30**, as shown in FIG. 2, depending on the direction which the wheelchair **34** faces when installed in the vehicle **20**. When the wheelchair **34** is forward-facing, the occupant of the wheelchair **34** faces the front end of the occupant compartment **30**. In other words, when the vehicle **20** is in forward motion along a roadway, the wheelchair **34** faces the direction of forward motion of the vehicle **20**. When the wheelchair **34** is rearward-facing, the occupant of the wheelchair **34** faces the rear end of the occupant compartment **30**. In other words, when the vehicle **20** is in forward motion along a roadway, the occupant faces the opposite direction of the forward motion of the vehicle **20**.

(18) The wheelchair **34** may be of any suitable type that supports a seated occupant and provides mobility to the seated occupant, i.e., the wheelchair **34** transports the seated occupant outside of the vehicle **20** and moves the seated occupant in the passenger compartment during ingress to and egress from the vehicle **20**. The wheelchair **34** supports the seated occupant in the occupant compartment **30** during operation of the vehicle **20**. The wheelchair **34** may include wheels that are latchable to the vehicle floor **22** as described further below.

(19) The vehicle **20** may include a plurality of latches **36** supported by the vehicle floor **22**. The latches **36** may engage wheels of the wheelchair **34** to inhibit movement of the wheels of the wheelchair **34** relative to the vehicle floor **22**, including along the vehicle-longitudinal axis **A1**. The latches **36** may be anchored to the vehicle floor **22** and may retractably extend through the wheels of the wheelchair **34**. The vehicle **20** may include any suitable number of latches **36** for the number of wheelchairs **34** that may be accommodated by the vehicle **20**. The latches **36** may be any suitable type of latches **36**. For example, the latches **36** may be of the type that are manually operated by an occupant of the wheelchair **34**. In other examples, the latches **36** may be of the type that are electronically operated, e.g., via a solenoid, to latch the wheels of the wheelchair **34**, e.g., through a human-machine interface such as a button, switch, touch-screen, etc. In the example in the Figures, the vehicle **20** includes six latches **36**. Four latches **36** are used in either direction that

the wheelchair **34** may be facing in the vehicle **20**.

(20) The vehicle **20** includes a track **38** for guiding movement of the post **24**, e.g., relative to the vehicle floor **22** of the vehicle **20**. The track **38** is at the vehicle floor **22**. The track **38** may be supported by, e.g., fixed to the vehicle floor **22**. The track **38** may be C-shaped in cross-section or have any other suitable shape for guiding movement of the post **24**, e.g., to permit movement of the post **24** along the track **38** and inhibit movement of the post **24** transverse to the track **38**.

(21) The track **38** may extend, at least partially, around the wheelchair seating area **32**. For example, as shown in FIGS. **1** and **2**, the track **38** may extend around the wheelchair seating area **32** on at least three sides of the wheelchair seating area **32**. As described further below, the track **38** extending around the wheelchair seating area **32** allows the post **24** and the support member **26** to move to one or more positions around the occupant depending on a direction the wheelchair **34** is facing.

(22) At least a portion of the track **38** may be vehicle-forward of the wheelchair seating area **32**, at least a portion of the track **38** may be vehicle-rearward of the wheelchair seating area **32**, and at least a portion may be vehicle-inboard of the wheelchair seating area **32**. For example, the track **38** may include a pair of cross-vehicle portions **40**, **42** and a vehicle-longitudinal portion **44**. The cross-vehicle portions **40**, **42** may extend along the cross-vehicle axis **A2**. The cross-vehicle portions **40**, **42** are spaced from each other along the vehicle-longitudinal axis **A1**. The vehicle-longitudinal portion **44** may extend along the vehicle-longitudinal axis **A1**, e.g., from one cross-vehicle portion **40**, **42** to the other cross-vehicle portion **40**, **42**. One of the cross-vehicle portions **40**, **42**, is vehicle-forward of the other of the cross-vehicle portions **40**, **42**, i.e., one of the cross-vehicle portions **40**, **42** is vehicle-rearward of the other of the cross-vehicle portions **40**, **42**. Specifically, the first cross-vehicle portion **40** is vehicle-forward of the second cross-vehicle portion **42**. Said differently, the second cross-vehicle portion **42** is vehicle-rearward of the first cross-vehicle portion **40**. The first cross-vehicle portion **40** may be vehicle-forward of the wheelchair seating area **32** and the second cross-vehicle portion **42** may be vehicle-rearward of the wheelchair seating area **32**. In other words, the wheelchair seating area **32** is between the first cross-vehicle portion **40** and the second cross-vehicle portion **42** along the vehicle-longitudinal axis **A1**.

(23) The cross-vehicle portions **40**, **42** are elongated at least partially along the cross-vehicle axis **A2**. In other words, the cross-vehicle portions **40**, **42** are longer along the cross-vehicle axis **A2** than along the vehicle-longitudinal axis **A1**. In some examples, the cross-vehicle portions **40**, **42** may extend entirely along, i.e., parallel with, the cross-vehicle axis **A2**. The cross-vehicle portions **40**, **42** each extend from a respective distal end **46**, **48** to the vehicle-longitudinal axis **A1**. In other words, the first cross-vehicle portion **40** extends from a first distal end **46** to the vehicle-longitudinal portion **44** and the second cross-vehicle portion **42** extends from a second distal end **48** to the vehicle-longitudinal portion **44**.

(24) The vehicle-longitudinal portion **44** is spaced vehicle-inboard from the distal ends **46**, **48** of the cross-vehicle portions **40**, **42** along the cross-vehicle axis **A2**. In other words, the vehicle-longitudinal portion **44** is spaced from the distal ends **46**, **48** of the cross-vehicle portions **40**, **42** toward a cross-vehicle center of the vehicle **20**. The vehicle-longitudinal portion **44** is vehicle-inboard of the wheelchair seating area **32**. The vehicle-longitudinal portion **44** is elongated along the vehicle-longitudinal axis **A1**. In other words, the vehicle-longitudinal portion **44** extends longer along the vehicle-longitudinal axis **A1** than along the cross-vehicle axis **A2**. The vehicle-longitudinal portion **44** may be elongated linearly along the vehicle-longitudinal axis **A1**. The vehicle-longitudinal portion **44** may be parallel with the vehicle-longitudinal axis **A1**. The vehicle-longitudinal portion **44** extends from one cross-vehicle portion **40**, **42** to the other cross-vehicle portion **40**, **42**. In other words, the vehicle-longitudinal portion **44** extends from the first cross-vehicle portion **40** to the second cross-vehicle portion **42**.

(25) The track **38** is continuous from the first distal end **46** of the first cross-vehicle portion **40** to

the second distal end **48** of the second cross-vehicle portion **42**. In other words, the track **38** is endless from the first distal end **46** of the first cross-vehicle portion **40** to the second distal end **48** of the second cross-vehicle portion **42**. The post **24** may slide along the track **38** from the first distal end **46** of the first cross-vehicle portion **40** to the second distal end **48** of the second cross-vehicle portion **42**.

(26) The vehicle **20** includes the post **24** to position and support the airbag **28**. The post **24** is supported by the vehicle floor **22**. In other words, weight of the post **24** is borne by the vehicle floor **22**. The post **24** may be elongated along a vertical axis **A4** and extend from the vehicle floor **22** toward the vehicle floor **22**. The vertical axis **A4** may be parallel with the vehicle-vertical axis **A3**. In the example shown in the Figures, the post **24** may be elongated from the vehicle floor **22** to a top end **50** spaced from the vehicle floor **22**. The post **24** may be of any suitable shape. For example, as shown in the Figures, the post **24** may be of a cylindrical shape.

(27) The post **24** is translatable relative to the vehicle floor **22**. For example, the post **24** may be slidably engaged with the track **38**, e.g., within a channel of the C-shaped cross-section as shown in FIG. **4**. The post **24** may be engaged with the track **38** in any suitable way such that the post **24** may be permitted to slide along the track **38** and inhibited from translating transverse to the track **38**. In the example shown in the Figures, the post **24** is slidable along the first cross-vehicle portion **40**, the second cross-vehicle portion **42**, and the vehicle-longitudinal portion **44** of the track **38**. In other words, the post **24** is slidable along the track **38** between the first distal end **46** and the second distal end **48**.

(28) The post **24** is translatable from a first position, shown in FIGS. **1** and **5**, to a second position, shown in FIGS. **2** and **6**. The post **24** at the first position may be at the first cross-vehicle portion **40** vehicle-forward of the wheelchair seating area **32**. The post **24** at the second position may be at the second cross-vehicle portion **42** vehicle-rearward of the wheelchair seating area **32**. The post **24** may be moved to the first position or the second position, e.g., in response to a command from a vehicle computer **52**. For example, and as further described below, the vehicle computer **52** may command the post **24** to the first position when a wheelchair **34** is secured vehicle-forward facing in the wheelchair seating area **32** and may command the post **24** to the second position when a wheelchair **34** is secured vehicle-rearward facing in the wheelchair seating area **32**.

(29) The vehicle **20** may include a first actuator **54** operatively coupled to the post **24** to move the post **24** along the track **38** from the first position to the second position, e.g., in response to a command from the vehicle computer **52**. The first and second positions are example positions, and the first actuator **54** may move the post **24** to various other positions along the track **38**, e.g., depending on a position of an occupant of the vehicle **20**. As shown in the FIG. **4**, the first actuator **54** may include a rack **56** extending along the track **38**, a pinion **58** engaged with the rack **56**, and a motor **60** operatively coupled to the pinion **58** to rotate the pinion **58**. The rack **56** may be fixed to the track **38**. Gear teeth of the pinion **58** may be meshed with gear teeth of the rack **56**. The motor **60** may be fixed to the post **24**. A drive shaft of the motor **60** may be coupled to the pinion **58** such that torque generated by the motor **60** is transmitted to the pinion **58** to move the post **24** along the track **38**. The first actuator **54** may include any other suitable mechanical, electromechanical, pneumatic, and/or hydraulic structure for moving the post **24** along the vehicle floor **22**, e.g., belts, pulleys, etc.

(30) The support member **26** is included to position and support the airbag **28**, e.g., in the uninflated position and the inflated position. The support member **26** is mounted to the post **24**, i.e., such that the support member **26** moves concurrently with the post **24**, e.g., along the track **38**. The support member **26** is rotatable relative to the post **24**, e.g., supported via bushing, bearing, or other suitable structure. The support member **26** is spaced from the vehicle floor **22**, e.g., at the top end **50** of the post **24**. For example, the support member **26** may be rotatably supported at the top end **50** of the post **24**. The support member **26** may be rotatable about the vertical axis **A4**. Rotation of the support member **26** about the vertical axis **A4** changes an azimuth angle of a facing direction of the airbag **28**, e.g., in a plane that extends through the cross-vehicle axis **A2** and the vehicle-

longitudinal axis A1 and to direct inflation of the airbag 28 toward an occupant of the vehicle 20.

(31) The support member 26 may be rotatable from a first orientation, shown in FIGS. 1 and 5, to a second orientation, shown in FIGS. 2 and 6. At the first orientation, the airbag 28 may face vehicle-rearward. At the second orientation, the airbag 28 may face vehicle-forward. The facing direction of the airbag 28 is a direction in which the airbag 28 generally moves when inflated from the uninflated position to the inflated position. The support member 26 may be at the first orientation when the post 24 is at the first position, e.g., to control kinematics of an occupant in a forward-facing wheelchair 34. The support member 26 may be at the second orientation when the post 24 is at the second position, e.g., to control kinematics of an occupant in a rearward-facing wheelchair 34. The support member 26 may be rotated to numerous orientations other than the first orientation and the second orientation. For example, the support member 26 may be oriented to the right or left of the first orientation and the second orientation, e.g., depending on a position along the cross-vehicle axis A2 of a wheelchair 34 and/or occupant within the vehicle 20.

(32) With reference to FIG. 3, the vehicle 20 may include a second actuator 62 that can generate force to rotate the support member 26. The second actuator 62 that may rotate the support member 26 to the first orientation, the second orientation, and/or various other orientations, e.g., depending on a position of an occupant of the vehicle 20. The second actuator 62 is operatively coupled to the support member 26 to rotate the support member 26 about the vertical axis A4, e.g., in response to a command from the vehicle computer 52. The second actuator 62 may be supported by the post 24. The second actuator 62 may be fixed to the post 24, e.g., via fastener or other suitable structure. For example, the second actuator 62 may include a motor. The motor may be supported by, e.g., fixed to the post 24. The motor may be internal of the post 24. A set of reduction gears may couple the motor to the support member 26 such that torque generated by the motor is provided to the support member 26. The second actuator 62 may include any other suitable mechanical, electromechanical, pneumatic, and/or hydraulic structure for rotating the support member 26 about the vertical axis A4, e.g., belts, pulleys, etc.

(33) The support member 26 may be rotatable about a horizontal axis A5. The horizontal axis A5 may be perpendicular to the vertical axis A4. Rotation of the support member 26 about the horizontal axis A5 changes an elevation angle the facing direction of the airbag 28, e.g., to angle the airbag 28 up or down. For example, the support member 26 may include a frame 64 and a base 66. The frame 64 may be mounted to the post 24 and rotate about the vertical axis A4. The base 66 may be supported by the frame 64. The base 66 may be rotatable about the horizontal axis A5 relative to the frame 64. For example, the base 66 may be rotatably supported by the frame 64 via bushings, bearings, etc. The airbag 28 may be fixed to the base 66.

(34) The vehicle 20 may include a third actuator 68 that can generate force to rotate the support member 26. The third actuator 68 may rotate the support member 26 to various orientations, e.g., depending on a position of an occupant of the vehicle 20. The third actuator 68 is operatively coupled to the support member 26 to rotate the support member 26 about the horizontal axis A5, e.g., in response to a command from the vehicle computer 52. The third actuator 68 may be supported by the post 24. In other words, weight of the actuator may be borne by the post 24. For example, the third actuator 68 may include a motor. The motor may be supported by, e.g., fixed to the frame 64 of the support member 26. The motor may be internal of the frame 64. A set of reduction gears may couple the motor to the base 66 such that torque generated by the motor is provided to the base 66 of the support member 26. The third actuator 68 may include any other suitable mechanical, electromechanical, pneumatic, and/or hydraulic structure for rotating the support member 26 about the horizontal axis A5, e.g., belts, pulleys, etc.

(35) The airbag 28 is included to control kinematics of an occupant of the vehicle 20. The airbag 28 is supported by the support member 26, e.g., fixed to the base 66 of the support member 26 via an airbag housing or other suitable structure. The airbag 28 may be fabric, e.g., a woven polymer. For

example, the fabric may be woven nylon yarn, for example, nylon 6, 6. Other examples of woven polymer include polyether ether ketone (PEEK), polyetherketoneketone (PEKK), polyester, etc. The woven polymer may include a coating, such as silicone, neoprene, urethane, etc. For example, the coating may be polyorgano siloxane. The airbag **28** may include one or more internal tethers (not shown) that maintain the position and shape of the airbag **28** in the inflated position.

(36) The airbag **28** is inflatable from the uninflated position, shown in Figures X-X, to the inflated position, shown in Figures X-X. The airbag **28** in the inflated position may control the kinematics of an occupant of the vehicle **20**. The airbag **28** may be inflated to the inflated position, e.g., by the vehicle computer **52**, in the event of certain vehicle impacts. A “certain impact” is an impact of the type and/or magnitude for which inflation of the airbag **28** is designed i.e., “certain” indicates the type and/or magnitude of the impact. The type and/or magnitude of such “certain impacts” may be pre-stored in the computer **52**, e.g., a restraints control module and/or a body control module. During such an event, the airbag **28** may inflate away from the support member **26** and toward the wheelchair seating area **32**. In other words, the airbag **28** may be between the support member **26** and the wheelchair seating area **32** when the airbag **28** is in the inflated position. For example, the airbag **28** in the inflated position may extend from the post **24** toward wheelchair seating area **32** when the post **24** is at both the first position and the second position. FIG. **5** shows the airbag **28** in the inflated position when the post **24** is in the first position at the first orientation. The airbag **28** faces vehicle-rearward when the support member **26** is at the first orientation, e.g. to control kinematics of an occupant of a vehicle-forward facing wheelchair **34**. FIG. **6** shows the airbag **28** in the inflated position when the post **24** is in the second position and the support member **26** is at the second orientation. The airbag **28** faces vehicle-forward when the support member **26** is at the second orientation, e.g. to control kinematics of an occupant of a vehicle-rearward facing wheelchair **34**.

(37) The support member **26** may include a covering (not numbered) to conceal the airbag **28** during normal operation of the vehicle **20**. In the event of certain vehicle impacts where the airbag **28** moves to the inflated position, a releasable seam (not shown) in the covering, e.g., a tear seam, may release to allow the airbag **28** to move to the inflated position.

(38) With reference to FIG. **7**, the vehicle **20** includes an inflator **70** that provides inflation medium to inflate the airbag **28** to the inflated position, e.g., in response to a command from the vehicle computer **52**. The inflator **70** is fluidly connected to the airbag **28**. The inflator **70** may be, for example, a pyrotechnic inflator that ignites a chemical reaction to generate the inflation medium, a stored gas inflator that releases (e.g., by a pyrotechnic valve) stored gas as the inflation medium, or a hybrid.

(39) The vehicle **20** may include one or more wheelchair position sensors **72** to identify the presence and a position of a wheelchair **34** in the vehicle **20**. The wheelchair position sensors **72** may be in communication with the vehicle computer **52** of the vehicle **20**. The wheelchair position sensors **72** may send a signal to the vehicle computer **52** to indicate that a wheelchair **34** is present in the vehicle **20** and the position of the wheelchair **34**. For example, the wheelchair position sensors **72** may indicate to the vehicle computer **52** whether the wheelchair **34** is forward-facing or rearward-facing. The wheelchair position sensors **72** may be any suitable type of sensors, e.g., weight sensors, cameras, contact switches, proximity sensors, etc.

(40) The vehicle **20** may include one or more occupant sensors **74** that detect characteristics of one or more occupants of the vehicle **20**, e.g., a presents, position, orientation, weight, and/or height of an occupant. The occupant sensors **74** may detect characteristics of specific body parts of an occupant, e.g., a position and/or orientation of a head of the occupant. The occupant sensors **74** may detect structures indicative of an occupant, e.g., the presence, position, and/or orientation of a wheelchair **34** in the wheelchair seating area **32**. For example, one or more image sensors may be supported within the passenger compartment and be orientated to face the occupant seating area. An image sensor could include one or more cameras, CCD image sensors, CMOS image sensors,

etc. As another example, proximity sensors, such as a capacitive proximity sensor, a photoelectric sensor, an infrared proximity sensor, etc., may be used to detect characteristics of one or more occupants of the vehicle **20**. The proximity sensors may detect, for example, the presence, the position, and/or the orientation of a wheelchair **34** in the wheelchair seating area **32**. As another example, the proximity sensors may detect a height of the occupant, a position of the occupant along the vehicle **20** longitudinal axis, etc. The occupant sensors **74** may include weight sensors, e.g., that detect a load applied to the vehicle floor **22** in the wheelchair seating area **32**. The occupant sensors **74** may include contact sensors that detect engagement of one or more latches **36** with a wheelchair **34**.

(41) The vehicle **20** may include at least one impact sensor **76** for sensing certain vehicle impacts. The impact sensor **76** is configured to detect certain vehicle impacts. A “certain vehicle impact” is an impact of the type and/or magnitude for which inflation of the airbag **28** is designed i.e., “certain” indicates the type and/or magnitude of the impact. The type and/or magnitude of such “certain vehicle impacts” may be pre-stored in the computer **52**, e.g., a restraints control module and/or a body control module. The impact sensor **76** may be of any suitable type, for example, post **24** contact sensors such as accelerometers, pressure sensors, and contact switches; and pre-impact sensors **76** such as radar, LIDAR, and vision sensing systems. The vision sensing systems may include one or more cameras, CCD image sensors, CMOS image sensors, etc. The impact sensor **76** may be located at numerous points in or on the vehicle **20**. The impact sensor **76** may be in communication with the vehicle computer **52**.

(42) The vehicle **20** may include a communication network **78**. The communication network **78** includes hardware, such as a communication bus, for facilitating communication among vehicle **20** components, e.g., the occupant sensor, the vehicle computer **52**, the impact sensors **76**, the inflator **70**, the first actuator **54**, the second actuator **62**, the third actuator **68**, etc. The communication network **78** may facilitate wired or wireless communication among the vehicle **20** components in accordance with a number of communication protocols such as controller area network (CAN), Ethernet, WiFi, Local Interconnect Network (LIN), and/or other wired or wireless mechanisms. Alternatively or additionally, in cases where the vehicle computer **52** comprises a plurality of devices, the communication network **78** may be used for communications between devices represented as the vehicle computer **52** in this disclosure

(43) With continued reference to FIG. 7, the vehicle computer **52** includes a processor and a memory storing instructions executable by the processor. The memory includes one or more forms of computer readable media, and stores instructions executable by the vehicle computer **52** for performing various operations, including as disclosed herein. The vehicle computer **52** may be a restraints control module. The vehicle computer **52** can be a generic computer with the processor and the memory as described above and/or may include an electronic control unit ECU or controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC (application specific integrated circuit) that is manufactured for a particular operation, e.g., an ASIC for processing sensor data and/or communicating the sensor data. In another example, the vehicle computer **52** may include an FPGA (Field-Programmable Gate Array) which is an integrated circuit manufactured to be configurable by a user. Typically, a hardware description language such as VHDL (Very High-Speed Integrated Circuit Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g. stored in a memory electrically connected to the FPGA circuit. In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in the vehicle computer **52**.

(44) The vehicle computer **52** may be programmed to, i.e., the memory may store executable by the processor to, detect a classification of an occupant of the vehicle **20**, i.e., an occupant classification.

The occupant classification is a certain detected characteristic of the occupant. For example, the occupant classification may include a certain detected occupant height, orientation of the occupant and/or the wheelchair **34** in the wheelchair seating area **32**, etc. The vehicle computer **52** may detect the occupant classification based on data received from the occupant sensors **74** and/or the wheelchair position sensors **72** e.g., via the communication network **78**. For example, data received from the wheelchair position sensors **72** may specify that the front two pairs of latches **36** are engaged with a wheelchair **34** and indicate that the occupant classification includes a forward-facing wheel chair, or the data received from the wheelchair position sensors **72** may specify that the rear two pairs of latches **36** are engaged with a wheelchair **34** and indicate that the occupant classification includes a rearward-facing wheel chair.

(45) The vehicle computer **52** may be programmed to command the first actuator **54** to move the post **24** along the track **38**, e.g., to the first position or the second position, based on a detected occupant classification. For example, the vehicle computer **52** may command the first actuator **54** to move the post **24** to first position when the detected occupant classification indicates that the occupant and/or the wheelchair **34** is facing vehicle-forward. The vehicle computer **52** may command the first actuator **54** to move the post **24** to second position when the detected occupant classification indicates that the occupant and/or the wheelchair **34** is facing vehicle-rearward. As another example, the vehicle computer **52** may store and use a look-up table to the like that associates various occupant classifications with various positions along the track **38** to command the first actuator **54**.

(46) The vehicle computer **52** may be programmed to command the second actuator **62** to rotate the support member **26** about the vertical axis **A4** based on a detected occupant classification. For example, the vehicle computer **52** may command the second actuator **62** to rotate the support member **26** to first orientation when the detected occupant classification indicates that the occupant and/or the wheelchair **34** is facing vehicle-forward. The vehicle computer **52** may command the second actuator **62** to rotate the post **24** to second orientation when the detected occupant classification indicates that the occupant and/or the wheelchair **34** is facing vehicle-rearward. As another example, the vehicle computer **52** may command the second actuator **62** to rotate the support member **26**, e.g., to the right or the left, such that the airbag **28** deploys toward a torso and/or head of the occupant when the detected occupant classification is a position of the occupant, a position specific part of the occupant, and/or a position of the wheelchair **34**. The vehicle computer **52** may store and use a look-up table to the like that associates various occupant classifications with various azimuth angles to command the second actuator **62**.

(47) The vehicle computer **52** may be programmed to command the third actuator **68** to rotate the support member **26** about the horizontal axis based on a detected occupant classification. For example, the vehicle computer **52** may command the third actuator **68** to rotate the support member **26**, e.g., increase or decrease the elevation angle, such that the airbag **28** deploys toward a head of the occupant when the detected occupant classification is a height of the occupant. The vehicle computer **52** may store and use a look-up table to the like that associates various occupant classifications with various elevation angles to command the third actuator **68**.

(48) FIG. **8** is a process flow diagram illustrating an exemplary process **800** for controlling inflation of the airbag **28**. The process **800** may be performed by the vehicle computer **52**. In other words, the memory of the vehicle computer **52** may store instructions executable by the processor to perform the process **800**. The process **800** begins in a block **805** in which the vehicle computer **52** collects data, e.g., received via the communication network **78** from the occupant sensor, the impact sensor **76**, etc. The vehicle computer **52** may continue to receive data throughout the process **800**. Throughout the process **800** in the present context means substantially continuously or at time intervals, e.g., every 100 milliseconds. The vehicle computer **52** may store the collected data, as well as other determinations, detections, commands, etc., received, performed, or transmitted throughout the process **800**.

(49) Next at a block **810** the vehicle computer **52** detects one or more occupant classifications, e.g., based on data received from the occupant sensors **74**. For example, the detected occupant classification may indicate that an occupant of the vehicle **20** is forward-facing or rearward-facing, that a wheelchair **34** in the wheelchair seating area **32** is forward-facing or rearward-facing, a height of an occupant, a position of an occupant along the cross-vehicle axis **A2**, etc.

(50) Next at a block **815** the vehicle computer **52** commands the first actuator **54**, the second actuator **62**, and/or the third actuator **68** to move the post **24** along the track **38** to a certain position, to rotate the support member **26** about the vertical axis **A4** to a certain orientation, and/or to rotate the support member **26** about the horizontal axis **A5** to a certain orientation based on the detected one or more occupant classifications, e.g., as described herein.

(51) At a block **820** the vehicle computer **52** determines whether a certain impact to the vehicle **20** has been detected based on data received from the impact sensors **76**. In response to determining that a certain impact has been detected, the vehicle computer **52** moves to a block **825**. In response to determining that a certain impact has not been detected, the vehicle computer **52** returns to the block **805**.

(52) At the block **825** the vehicle computer **52** inflates the airbag **28** to the inflated position, e.g., by sending a command, such as an electrical pulse, to the inflator **70** via the communication network **78**. After the **825** the process **800** may end.

(53) In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

(54) Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random-access memory, etc.

(55) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a

computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

(56) Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

(57) In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

(58) With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. Operations, systems, and methods described herein should always be implemented and/or performed in accordance with an applicable owner's/user's manual and/or safety guidelines.

(59) The numerical adjectives “first,” “second,” etc., are used herein merely as identifiers and do not signify order or importance. Use of “in response to,” “based on,” and “upon” herein indicates a causal relationship, not merely a temporal relationship.

(60) The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A vehicle comprising: a vehicle floor; a post supported by the vehicle floor, the post being translatable relative to the vehicle floor; a support member mounted to the post, the support member being rotatable relative to the post about a horizontal axis; and an airbag supported by the support member.
2. The vehicle of claim 1, wherein the support member is rotatable from a first orientation in which the airbag faces vehicle-rearward to a second orientation in which the airbag faces vehicle-forward.
3. The vehicle of claim 1, wherein the support member is rotatable about a vertical axis.
4. The vehicle of claim 3, further comprising an actuator operatively coupled to the support member to rotate the support member about the vertical axis.
5. The vehicle of claim 4, wherein the actuator is supported by the post.
6. The vehicle of claim 4, further comprising a computer having a processor and a memory storing instructions executable by the processor to command the actuator to rotate the support member about the vertical axis based on a detected occupant classification.
7. The vehicle of claim 1, further comprising an actuator operatively coupled to the support

member to rotate the support member about the horizontal axis.

8. The vehicle of claim 7, further comprising a computer having a processor and a memory storing instructions executable by the processor to command the actuator to rotate the support member about the horizontal axis based on a detected occupant classification.

9. The vehicle of claim 8, wherein the detected occupant classification includes a detected occupant height.

10. The vehicle of claim 7, wherein the actuator is supported by the post.

11. The vehicle of claim 1, further comprising a track at the vehicle floor, the post being slidably engaged with the track.

12. The vehicle of claim 11, further comprising an actuator operatively coupled to the post to move the post along the track.

13. The vehicle of claim 12, further comprising a computer having a processor and a memory storing instructions executable by the processor to command the actuator to move the post along the track to a first position or a second position based on a detected occupant classification.

14. The vehicle of claim 12, wherein the actuator includes a rack extending along the track, a pinion engaged with the rack, and a motor operatively coupled to the pinion to rotate the pinion.

15. The vehicle of claim 11, wherein the track includes a pair of cross-vehicle portions and a vehicle-longitudinal portion extending from one cross-vehicle portion of the pair of cross-vehicle portions to the other cross-vehicle portion of the pair of cross-vehicle portions, the pair of cross-vehicle portions extending along a cross-vehicle axis and the vehicle-longitudinal portion extending along a vehicle-longitudinal axis.

16. The vehicle of claim 15, further comprising a wheelchair seating area between the cross-vehicle portions along the vehicle-longitudinal axis.

17. The vehicle of claim 11, further comprising a wheelchair seating area, the track extending around the wheelchair seating area on at least three sides of the wheelchair seating area.

18. A vehicle comprising: a vehicle floor; a post supported by the vehicle floor, the post being translatable relative to the vehicle floor; a support member mounted to the post, the support member being rotatable relative to the post; an airbag supported by the support member; and a track at the vehicle floor, the post being slidably engaged with the track; the track including a pair of cross-vehicle portions and a vehicle-longitudinal portion extending from one cross-vehicle portion of the pair of cross-vehicle portions to the other cross-vehicle portion of the pair of cross-vehicle portions, the pair of cross-vehicle portions extending along a cross-vehicle axis and the vehicle-longitudinal portion extending along a vehicle-longitudinal axis.

19. The vehicle of claim 18, further comprising a wheelchair seating area between the cross-vehicle portions along the vehicle-longitudinal axis.

20. The vehicle of claim 18, further comprising: an actuator operatively coupled to the support member to rotate the support member relative to the post about a vertical axis and a horizontal axis; a computer having a processor and a memory storing instructions executable by the processor to command the actuator to rotate the support member about the vertical axis and the horizontal axis based on a detected occupant classification.
